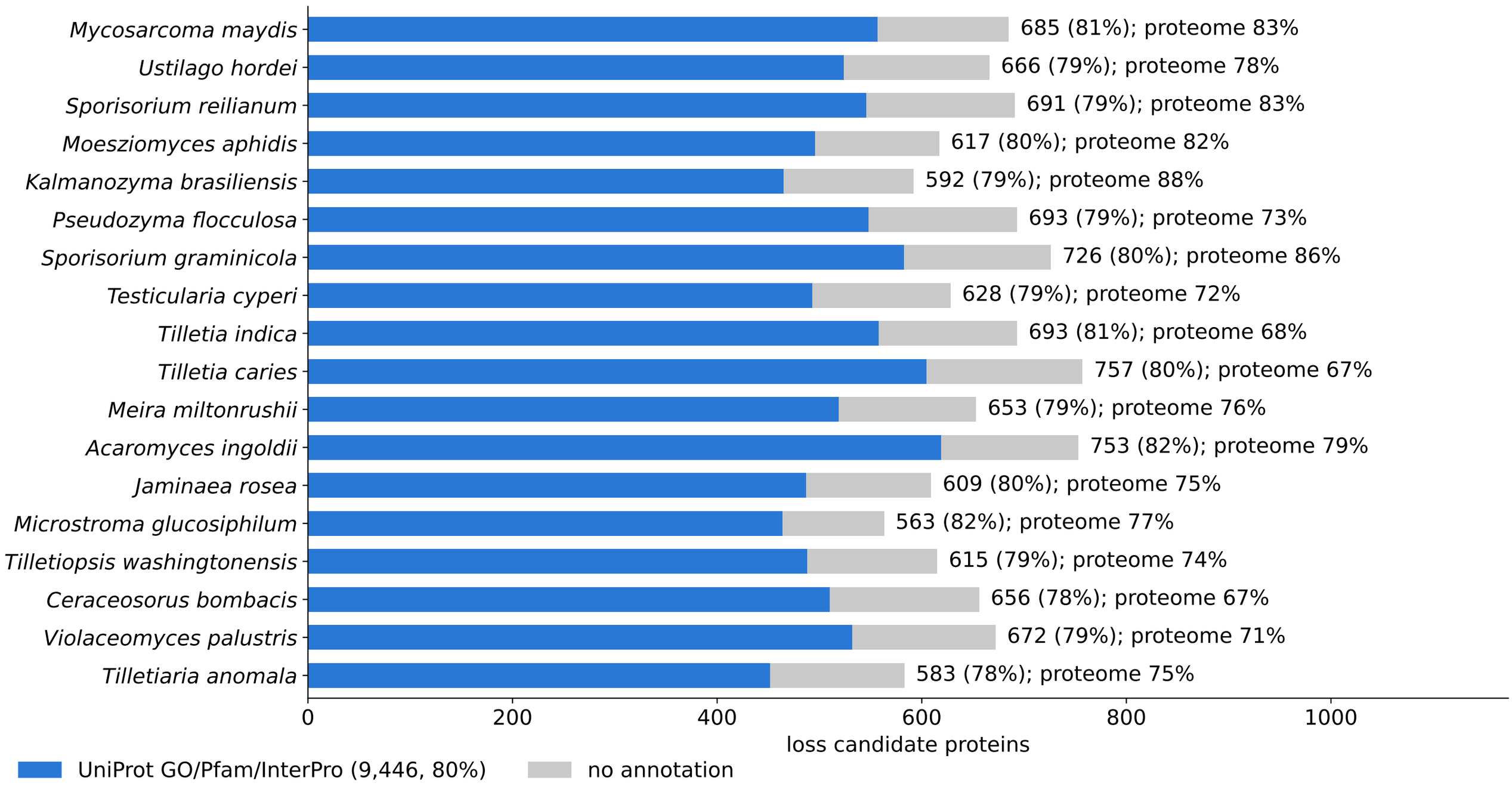
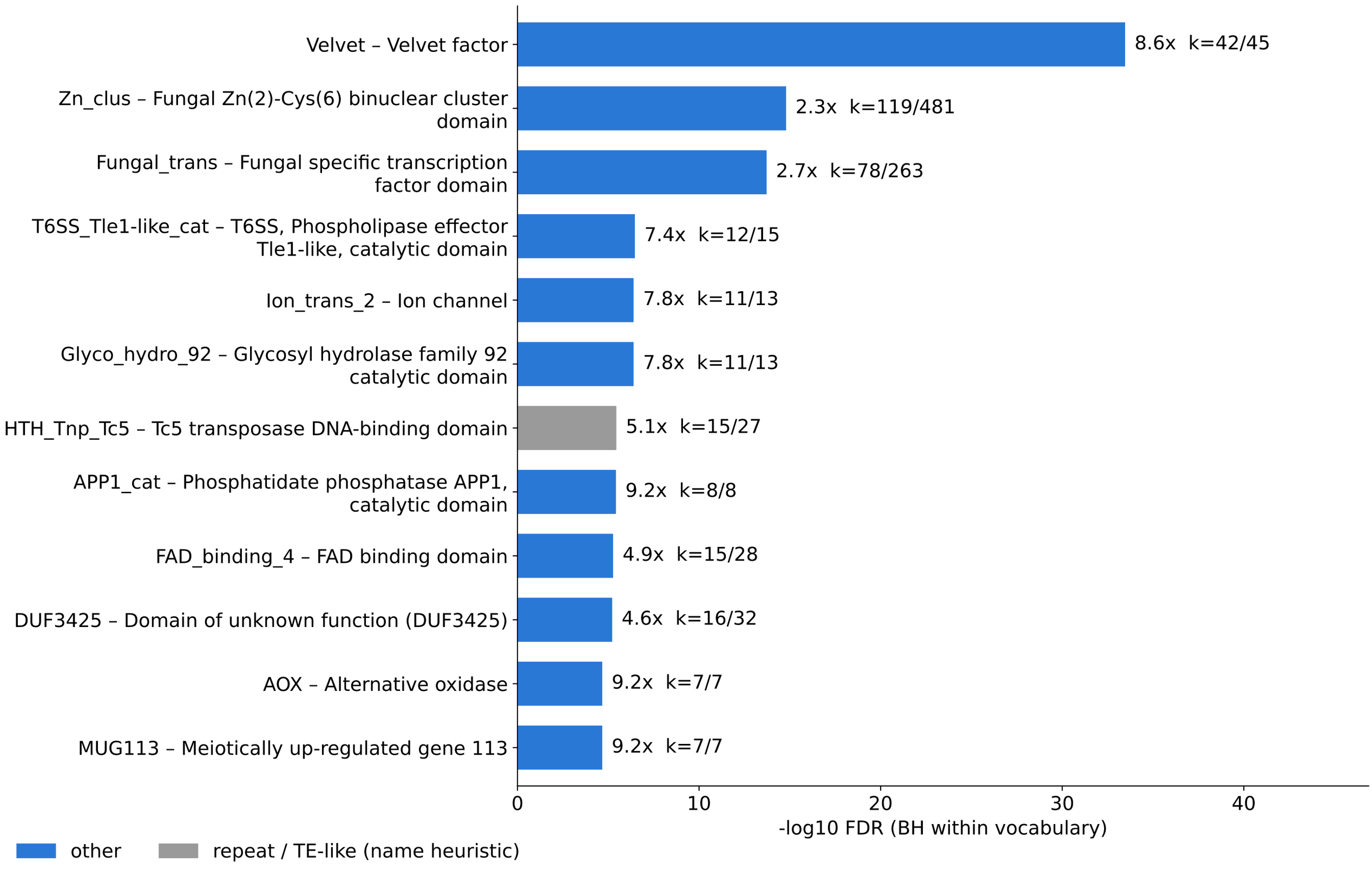
Ustilaginomycotina (18 OUT) vs Malassezia (10 IN): enrichment of loss-direction candidates
(present in >=0.75 of OUT, absent from all IN; diamond pairwise, nf_NovInvenio run 2026-10-11)

A 11,852 candidates in 18 OUT species



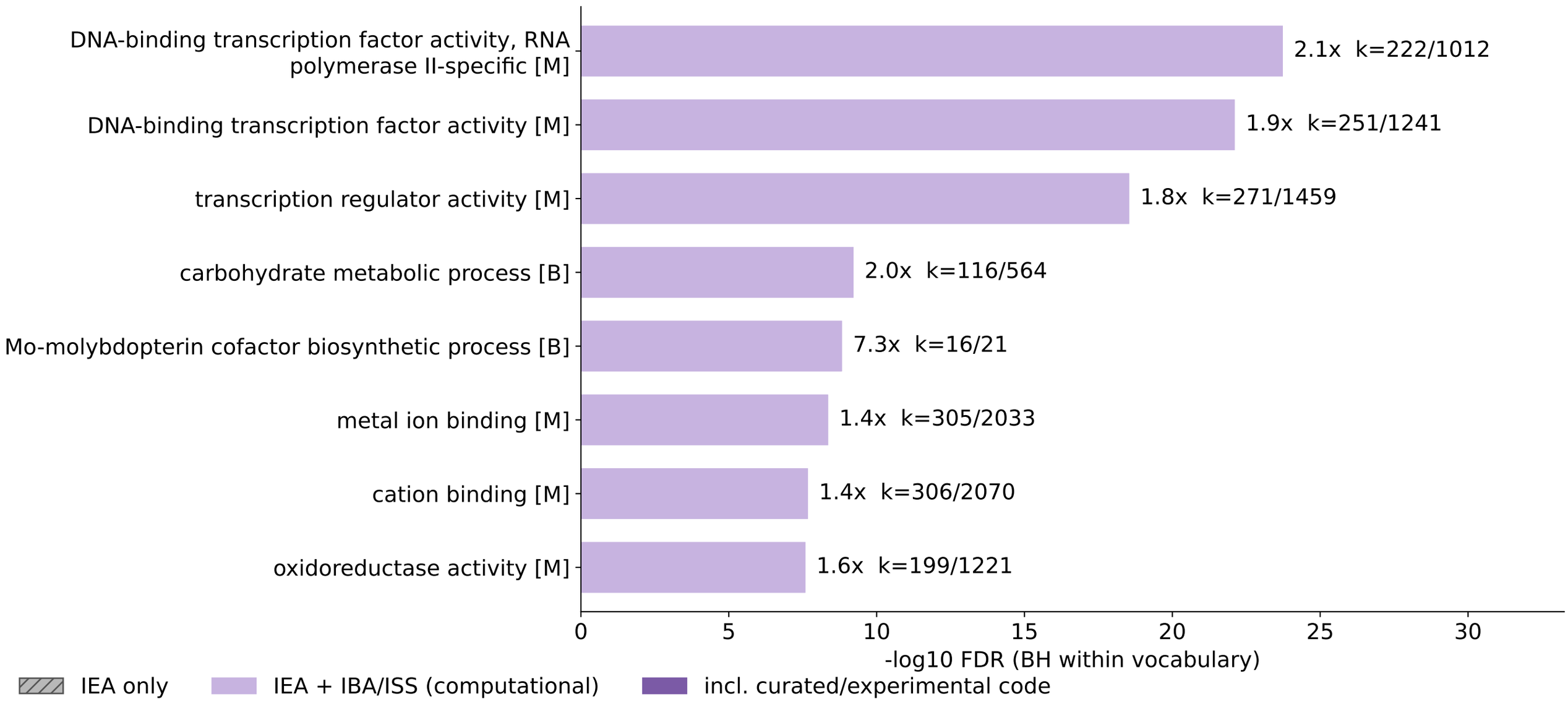
B Pfam families enriched in candidates (primary; family units on both sides)

1527 terms tested; 125 at FDR<0.05; annotated units: 2051 cand. / 18903 all



C GO terms (secondary; UniProt computational evidence)

4032 terms tested; 207 at FDR<0.05; annotated units: 1842 cand. / 17553 all



Annotation: UniProt reference proteomes (Pfam, InterPro, GO). Universe: all proteins of the 18 OUT proteomes. Family units: loss-candidate families from the pipeline (mmseqs on loss_candidates.fa); background clustered with mmseqs (0.3 id, 0.8 cov, cov-mode 0). One-sided hypergeometric; BH within each vocabulary. Bars: FDR<0.05 and k>=3 candidate units. Loss candidates: 11852; annotated fraction 0.797.